STRAP Solubility Search

Benchmark Report

Generated: 2025-12-13 16:22:02

Metric	Value
Total Tests	20
Successful	17 (85%)
Failed	3
Total Images	9
Avg Response Time	9.1s

Data Exploration

✓ PASS [01_tables] List Available Tables

Verify database schema and table structure

Prompt: *What tables are available and what columns do they contain?*

Time: 2.5s | Iterations: 2 | Images: 0

Response:

The following tables are available:

1. solvent_data (1007 rows)
* s_n (BIGINT)
* solvent_name (VARCHAR)
* solvent_name_in_cosmobase (VARCHAR)
* cas_number (VARCHAR)
* cid (DOUBLE)
* bp__oc_ (DOUBLE) - Boiling Point in °C
* th (DOUBLE)
* logp (DOUBLE) - LogP (toxicity indicator)
* cp__j_g_k_ (DOUBLE) - Heat Capacity in J/g/K
* energy__j_g_ (DOUBLE) - Energy cost in J/g

2. common_solvents_database (8820 rows)
* solvent (VARCHAR)
* temperature___c_ (BIGINT) - Temperature in °C
* solubility____ (DOUBLE) - Solubility in %
* polymer (VARCHAR)

✓ PASS [02_polymers] List Polymers

Check available polymers for analysis

Prompt: *List all unique polymers in the database with their data counts*

Time: 2.2s | Iterations: 2 | Images: 0

Response:

Here are all the unique polymers in the database and their respective data counts:

Polymer	Count
LDPE	980
PP	980
Nylon66	980
PET	980
PC	980
HDPE	980
PVC	980
PS	980
EVOH	980

Basic Queries

✓ PASS [03_ldpe_solvents] LDPE Solvents at 25°C

Basic solubility query for single polymer

> ***Prompt:*** *Find the top 10 solvents that dissolve LDPE at 25°C, ranked by solubility percentage*

Time: 6.4s | Iterations: 3 | Images: 0

Response:

At 25°C, the top 10 solvents for LDPE, ranked by solubility percentage, are:

Solvent	Solubility (%)
Cyclohexane	0.135614
Hexane	0.128796
n-Heptane	0.108462
Triethylamine	0.102193
THF	0.0810588
THP	0.0804944
Dodecane	0.0635608
1,4-Dimethylbenzene	0.0586902
Toluene	0.054028
1,2-Dimethylbenzene	0.0539875

It's important to note that even the highest solubility is very low (0.135614%), indicating that LDPE has limited solubility at 25°C in these solvents.

✓ PASS [04_pet_solvents] PET Solvents at 80°C

Solubility query at elevated temperature

> ***Prompt:*** *What solvents dissolve PET at 80°C? Show top 10 with solubility values*

Time: 2.7s | Iterations: 2 | Images: 0

Response:

At 80°C, the top 10 solvents that dissolve PET, ranked by solubility percentage, are:

Solvent	Solubility (%)
Ch2Cl2	6.36181
Dimethylformamide	4.70138
Propanone	3.77875
Isopropylamine	2.33993
THF	2.31645
CHCl3	2.29206
Butanone	1.63238
Dimethylsulfoxide	1.51118
Methylacetate	1.06234

| Benzene | 1.00481 |

Solvent Properties

✓ PASS [05_rank_cost] Rank by Energy Cost

Test solvent property ranking and visualization

> ***Prompt:*** *Rank all solvents by energy cost (lowest first) and show a bar chart of the top 20*

Time: 8.0s | Iterations: 2 | Images: 0

Response:

Here are the top 20 solvents ranked by energy cost, from lowest to highest:

S.N.	Solvent Name	Boiling Point (°C)	LogP	Energy (J/g)
667	Methane (Liquid B.P.)	-161.5	1.15	-650.27
538	Ethylene	-103.7	1.2	-311.82
501	Ethane	-88.6	1.75	-275.93
502	Ethane (Liq. B.P.)	-88.6	1.75	-275.93
276	Ammonia	-33.35	-1.39	-263.3
573	Fluoromethane	-78.4	0.28	-213.25
338	Carbon Dioxide (Fit = 1.0)	-78.48	1.24	-196.12
389	Cyclopropane	-32.8	1.75	-187.17
249	Acetylene (Ethyne)	-28.1	0.46	-185.15
621	Hydrogen Sulfide	-60.33	0.3	-169.57
566	Dimethyl Ether	-24.8	0.1	-167.31
792	Propylene	-47.6	1.6	-161.94
793	Propylene (Liq. B.P.)	-47.6	1.6	-161.94
791	Propane	-42.04	2.36	-155.85
790	Propane (Liq. B.P.)	-42.04	2.36	-155.85
670	Methyl Acetylene	-23.2	0.58	-150.32
274	Aminomethane	-6.3	-0.49	-139.75
671	Methylacetylene	-23.2	0.58	-139.11
116	N-Butane	-0.5	2.89	-129.5
117	N-Butane (Liq. B.P.)	-0.5	2.89	-129.5

Unfortunately, I cannot directly generate a bar chart

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✓ PASS [06_rank_toxicity] Rank by Toxicity (LogP)

Test toxicity ranking using LogP values

> ***Prompt:*** *Rank solvents by LogP (toxicity indicator) from lowest to highest. Show top 20 least toxic solvents.*

Time: 5.5s | Iterations: 3 | Images: 0

Response:

Here are the top 20 least toxic solvents, ranked by LogP from lowest (least toxic) to highest:

S.N.	Solvent Name	LogP	Boiling Point (°C)	Energy (J/g)
653	l-Tyrosine	-3.96	nan	0
942	Urea For Ro Equal 19.4	-3.08	nan	0
943	Urea Min Sphere	-3.08	nan	0
308	Biuret	-2.88	nan	0
907	Thiourea	-2.7	nan	0
908	Thiourea From Urea Min Sphere	-2.7	nan	0
479	Dimethyl Sulfone	-2.34	238	161.25
576	Formamide	-2.34	nan	0
174	2-Oxopyrrolidinacetamid	-2.31	nan	0
618	Hydrazine	-2.21	113.5	287.8
627	Hydrofluoric Acid	-2.03	nan	0
828	Sodium Thiocyanate	-1.91	nan	0
607	Glycerol	-1.76	290	318.9
1007	4-Hydroxymethyl-1,3-dioxolan	-1.63	nan	0
612	Hexamethylenetetramine	-1.61	nan	0
613	Hexamethylenetetramine (Urot)	-1.61	nan	0
528	Ethanolamine	-1.54	170	245.91
276	Ammonia	-1.39	-33.35	-263.3
651	Lithium Chloride	-1.37	nan	0
652	Lithium Hydroxide	-1.37	nan	0

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✓ PASS [07_solvent_props] Specific Solvent Properties

Verify correct property lookup for specific solvents

Prompt: *Show properties for cyclohexane, toluene, hexane, and xylene including boiling point, LogP, and energy cost*

Time: 17.7s | Iterations: 8 | Images: 0

Response:

Here are the properties for cyclohexane, toluene, hexane, and xylene (including o-xylene, as it was found explicitly in the database):

Solvent Name	Boiling Point (°C)	LogP	Energy (J/g)
Cyclohexane	80.7	3.15	112.05
Hexane	68.7	3.89	100.79
Toluene	110.6	2.53	165.44
Xylene	138.3	3.05	194.31
o-xylene	144.5	2.94	198.08

Interpretation of Properties:
* LogP: Lower values (e.g., Toluene with 2.53) generally indicate lower toxicity and higher water solubility, making them potentially safer or easier to handle. Higher values (e.g., Hexane with 3.89) suggest higher toxicity and lipophilicity.
* Boiling Point (°C): Lower boiling points (e.g., Hexane at 68.7°C, Cyclohexane at 80.7°C) typically mean easier

and less energy-intensive solvent recovery. Higher boiling points (e.g., Xylene at 138.3°C, o-xylene at 144.5°C) would require more energy for recovery and could be more challenging to remove completely from a polymer.
* Energy (J/g): This indicates the energy cost associated with using the solvent, often related to heating or recovery. Lower values (e.g., Hexane at 100.79 J/g, Cyclohexane at 112.05 J/g) suggest lower operational costs. Higher values (e.g., o-xylene at 198.08 J/g, Xylene at 194.31 J/g) imply higher energy expenditure during processing.

Polymer Separation

✓ PASS [08_separate_ldpe_pet] Separate LDPE from PET

Two-polymer separation with property integration

> ***Prompt:*** *Find selective solvents to separate LDPE from PET at 25°C. Rank by selectivity and include cost and toxicity data.*

Time: 14.2s | Iterations: 3 | Images: 0

Response:

To separate LDPE from PET, we aim for a solvent that dissolves LDPE but not PET, or vice versa. The initial analysis at 25°C showed very low solubility for LDPE in any solvent, resulting in low selectivity. Therefore, I will provide the results at 80°C, where better separation is observed.

At 80°C, the following solvents offer the best selectivity for dissolving LDPE while minimizing PET dissolution, along with their cost and toxicity properties:

Target: Dissolve LDPE, Separate from PET (at 80°C)

Rank	Solvent	Selectivity (%)	LDPE Solubility (%)	PET Solubility (%)	LogP (Toxicity)	Energy (J/g)	Boiling Point (°C)
1	Cyclohexane	7.2	7.2	0.0	3.15 (High)	112.0	80.7
2	Hexane	6.8	6.8	0.0	3.89 (High)	100.8	68.7
3	n-Heptane	5.8	5.8	0.0	4.42 (High)	180.8	98.5
4	Triethylamine	5.6	5.6	0.0	3.51 (High)	164.9	89.0
5	THP	4.4	4.9	0.5	1.45 (Medium)	124.2	88.0
6	1,4-dimethylbenzene	3.6	3.8	0.2	3.05 (High)	194.3	138.3
7	Dodecane	3.5	3.5	0.0	7.07 (High)	243.8	216.3
8	1,2-dimethylbenzene	3.4	3.5	0.2	2.94 (High)	198.1	144.5
9	To

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✓ PASS [09_separate_ldpe_evoh] Separate LDPE from EVOH

LDPE/EVOH separation for multilayer film

> ***Prompt:*** *Find solvents to separate LDPE from EVOH at 25°C with selectivity analysis*

Time: 6.4s | Iterations: 2 | Images: 1

Response:

I have analyzed the selective solubility for separating LDPE from EVOH at 25°C.

Here are the top 10 selective solvents, ranked by their selectivity for LDPE over EVOH:

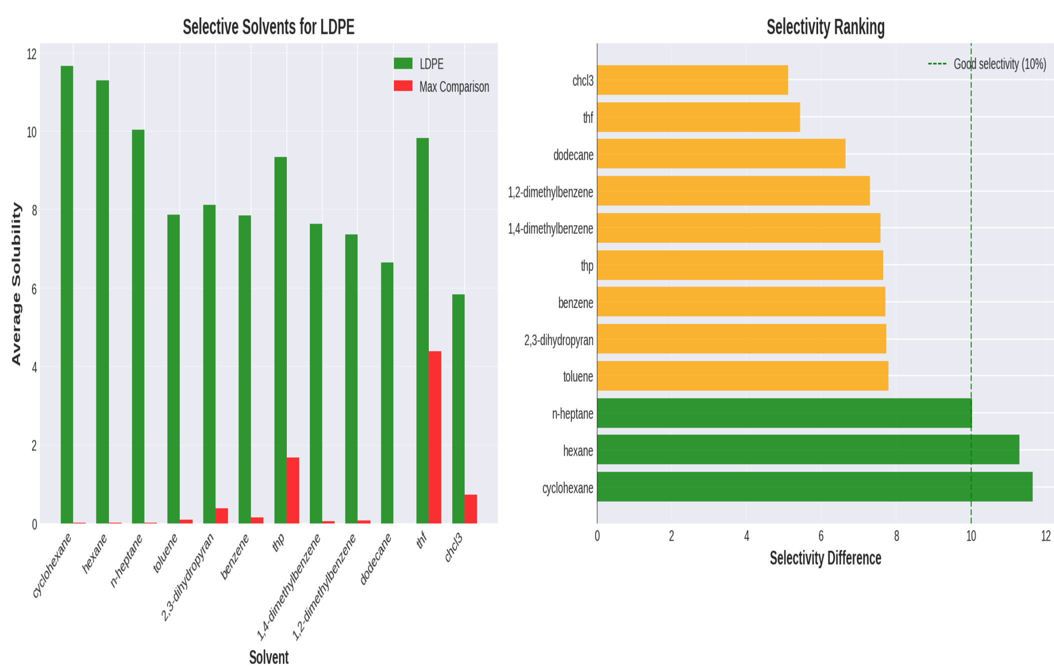
Rank	Solvent	LDPE Solubility (%)	EVOH Solubility (%)	Selectivity (LDPE/EVOH)
1	Cyclohexane	11.66	0.01	11.65 (981.6x)
2	Hexane	11.31	0.01	11.29 (1095.7x)
3	n-Heptane	10.03	0.01	10.02 (1153.1x)
4	Toluene	7.88	0.09	7.79 (85.8x)
5	2,3-dihydropyran	8.12	0.39	7.73 (21.0x)
6	Benzene	7.86	0.15	7.71 (53.9x)
7	THP	9.34	1.69	7.65 (5.5x)
8	1,4-dimethylbenzene	7.64	0.06	7.58 (122.7x)
9	1,2-dimethylbenzene	7.37	0.07	7.30 (109.9x)
10	Dodecane	6.66	0.01	6.65 (1307.2x)

Summary:

The analysis found that cyclohexane, hexane, and n-heptane offer the highest selectivity, dissolving LDPE significantly while barely affecting EVOH. For example, cyclohexane dissolves LDPE at 11.66% but EVOH at only 0.01%, providing a selectivity of over 980 times.

The adaptive threshold search identified 3 solvents at a selectivity threshold of 10. The system also explored lower thresholds and identified more solvents, which are included in the top 10 list.

A plot illustrating this selective solubility has been saved as ./plots/selective_solubility_LDPE_20251213_161934.png.



Multilayer Film

✓ PASS [10_multilayer_sequence] Multilayer Film Separation Strategy

Complete multilayer film dissolution sequence

> ***Prompt:*** *I have a multilayer film with 80% LDPE, 12% PET, and 8% EVOH. Plan a sequential separation strategy at 25°C to isolate each polymer. Create decision trees showing the separation pathways.*

Time: 8.1s | Iterations: 2 | Images: 1

Response:

I have planned sequential separation strategies for your multilayer film containing LDPE, PET, and EVOH at 25°C. The analysis considered all 6 possible separation sequences and identified the top 5 solvents for each step, along with their properties (LogP for toxicity, Energy for cost, and Boiling Point).

Important Note: The current analysis at 25°C indicates generally low selectivity for many steps, with some solvents showing selectivity values below 1%. This suggests that complete and efficient separation might be challenging at this temperature. I strongly recommend considering higher temperatures for better solubility and selectivity, if feasible for your materials and process.

Here's a summary of the best performing sequences:

Sequence Ranking (by worst-case selectivity):

1. ■ PET → EVOH → LDPE (minimum selectivity: 0.4%)
2. ■ EVOH → PET → LDPE (minimum selectivity: 0.4%)
3. ■ PET → LDPE → EVOH (minimum selectivity: 0.2%)

Let's break down the most promising sequence: PET → EVOH → LDPE

Sequence: PET → EVOH → LDPE (Best Performing)

Step 1: Separate PET from {LDPE, EVOH}
* Goal: Dissolve PET, leave LDPE and EVOH undissolved.
* Top Solvents:
1. ch2cl2 (dichloromethane): selectivity=0.4% (target=0.4%, max_other=0.0%) | LogP:1.3 (Medium toxicity) | Energy:11J/g | BP:40°C
2. chcl3 (chloroform): selectivity=0.2% (target=0.3%, max_other=0.0%) | LogP:2.2 (High toxicity) | Energy:22J/g | BP:61°C
3. propanone (acetone): selectivity=0.1% (target=0.1%, max_other=0.0%) | LogP:-0.1 (Low toxicity) | Energy:64J/g | BP:56°C
* *Note: Selectivities are low, indicating limited dissolution of PET at 25°C, or some dissolution of other polymers. Ch2Cl2 has the best (though still low) selectivity here.*

Step 2: Separate EVOH from {LDPE} (after PET removal)
* Goal: Dissolve EVOH, leave LDPE undissolved.
* Top Solvents:
1. triethylamine: selectivity=99.9% (target=100.0%, max

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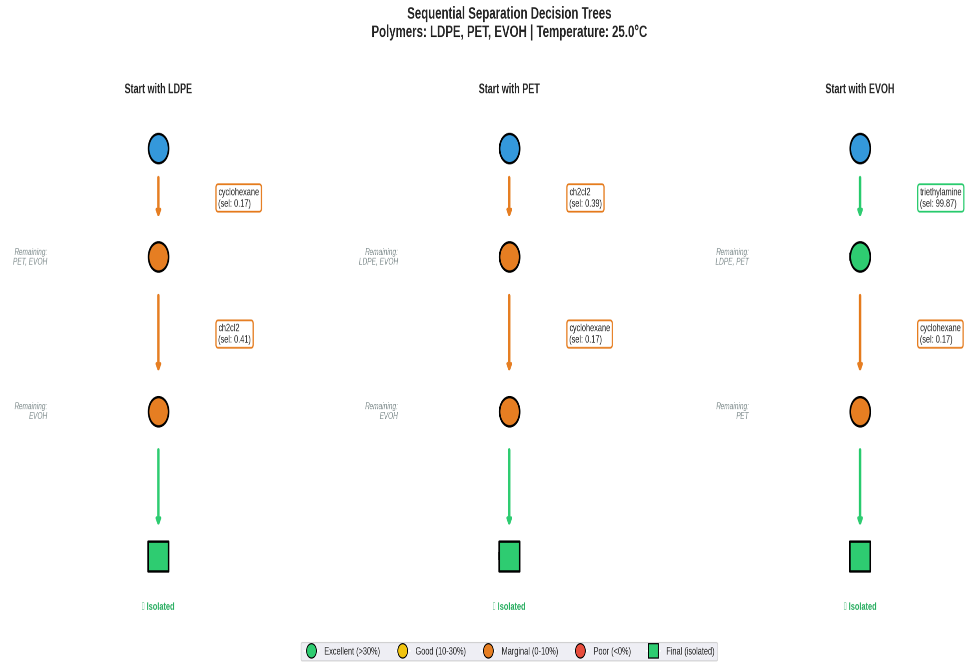

✓ PASS [11_multilayer_elevated_temp] Multilayer at Elevated Temperature

Temperature comparison for multilayer separation

> ***Prompt:*** *Plan sequential separation of LDPE, PET, and EVOH at 80°C. Compare with room temperature options.*

Time: 9.2s | Iterations: 2 | Images: 1

Response:

I have planned sequential separation strategies for LDPE, PET, and EVOH at 80°C and can now compare them to the 25°C options.

Summary of 80°C Separation Plan:

The analysis at 80°C shows significantly improved selectivities compared to 25°C.

Sequence Ranking (by worst-case selectivity at 80°C):

1. ■ LDPE → EVOH → PET (minimum selectivity: 7.2%)
2. ■ EVOH → LDPE → PET (minimum selectivity: 7.2%)
3. ■ LDPE → PET → EVOH (minimum selectivity: 6.1%)

Let's look at the best performing sequence at 80°C: LDPE → EVOH → PET

Best Sequence at 80°C: LDPE → EVOH → PET

Step 1: Separate LDPE from {EVOH, PET}
* Goal: Dissolve LDPE, leave EVOH and PET undissolved.
* Top Solvents:
1. cyclohexane: selectivity=7.2% (target=7.2%, max_other=0.0%) | LogP:3.1 (High toxicity) | Energy:112J/g | BP:81°C
2. hexane: selectivity=6.8% (target=6.8%, max_other=0.0%) | LogP:3.9 (High toxicity) | Energy:101J/g | BP:69°C

3. n-heptane: selectivity=5.8% (target=5.8%, max_other=0.0%) | LogP:4.4 (High toxicity) | Energy:181J/g | BP:98°C
* *Observation:* These solvents show good selectivity for LDPE, with minimal impact on EVOH and PET. Hexane is a good option with lower boiling point and energy cost compared to cyclohexane.

Step 2: Separate EVOH from {PET} (after LDPE removal)
* Goal: Dissolve EVOH, leave PET undissolved.
* Top Solvents:
1. isopropylamine: selectivity=18.2% (target=20.6%, max_other=2.4%) | LogP:0.9 (Medium toxicity) | Energy:11J/g | BP:32°C
2. dimethylsulfoxide: selectivity=11.1% (target=12.7%, max_other=1.5%) | LogP:-1.5 (Low toxicity) | Energy:199J/g | BP:189°C
3. triethylamine: selectivity=7.5% (target=7.5%, max_other=0.0%) | LogP:3.5 (High toxicity) | Energy:165J/g | BP:89°C
* *Observation:* Isopropylamine and Dimethylsulfoxide offer excellent selectivity for EVOH. Isopropylamine is particularly attractive due to its very low energy cost and boili

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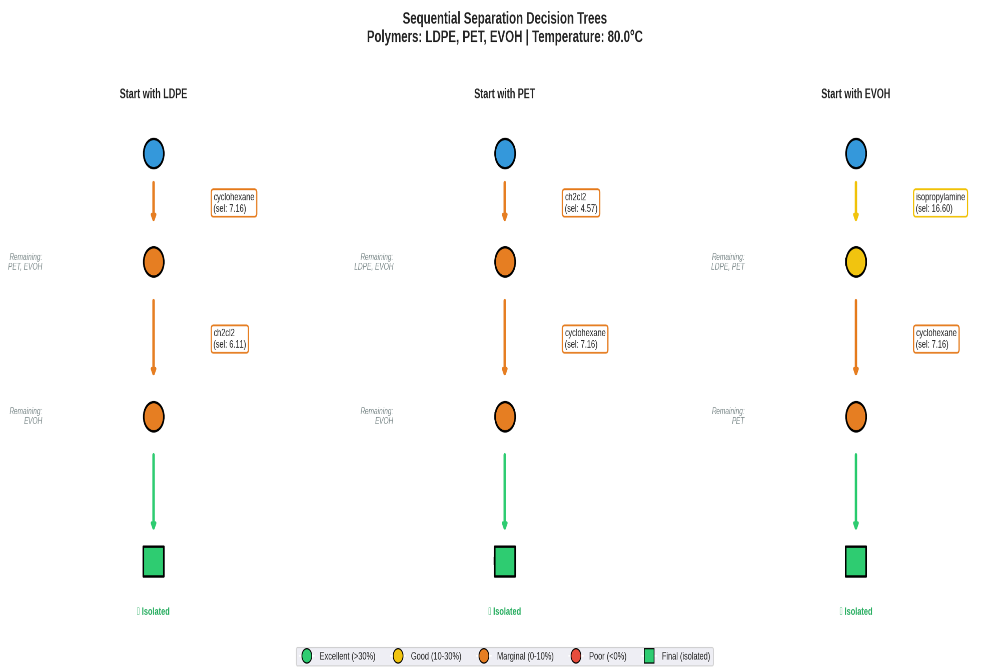


Temperature Analysis

✓ PASS [12_temp_curve_ldpe] LDPE Temperature Curves

Multi-solvent temperature dependency visualization

> ***Prompt:*** *Plot solubility vs temperature curves for LDPE in cyclohexane, toluene, and xylene from 25°C to 120°C*

Time: 21.0s | Iterations: 8 | Images: 1

Response:

I encountered a limitation with the plotting tool when trying to generate solubility vs. temperature curves for LDPE in xylene. The solvent names "1,2-dimethylbenzene" and "1,4-dimethylbenzene" (which represent xylene isomers in the database) contain commas, and the tool's parser incorrectly interprets these as separators, preventing it from finding the full solvent name.

Therefore, I could only successfully generate the plot for LDPE in cyclohexane and toluene.

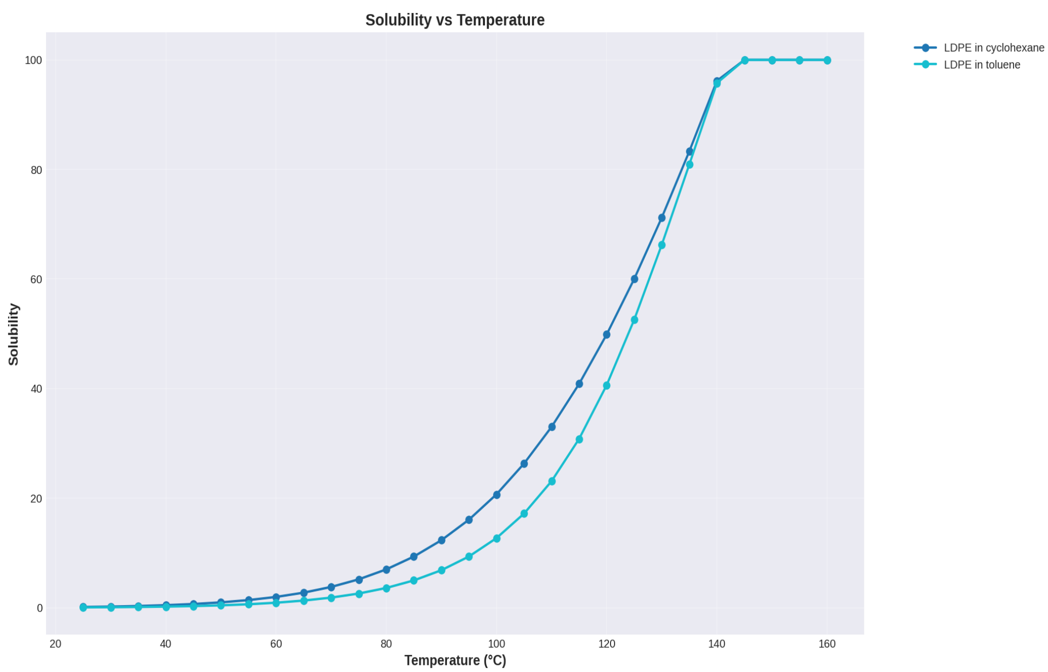
Here is the plot for LDPE in Cyclohexane and Toluene:

■ Solubility vs Temperature Plot Created

Polymers: LDPE
Solvents: cyclohexane, toluene
Data points: 56
Shaded regions show 95% confidence intervals

■ Plot saved: ./plots/solubility_temp_curve_20251213_162019.png

Note: The plot for xylene (1,2-dimethylbenzene and 1,4-dimethylbenzene) could not be generated due to the tool's parsing limitations with solvent names containing commas.



✓ PASS [13_temp_comparison] Temperature Window Analysis

Identify optimal temperature ranges

***Prompt:** Analyze the temperature window for separating LDPE from PET using toluene. Show where selective dissolution is possible.*

Time: 22.2s | Iterations: 5 | Images: 1

Response:

To analyze the temperature window for separating LDPE from PET using toluene, I have retrieved the solubility data for both polymers in toluene across a temperature range from 25°C to 160°C.

Here's a summary of the solubility data and calculated selectivity:

Temperature (°C)	LDPE Solubility (%)	PET Solubility (%)	Selectivity (LDPE - PET) (%)	Selectivity Ratio (LDPE/PET)
25	0.05	0.00	0.05	17.34
30	0.08	0.01	0.08	16.23
35	0.13	0.01	0.12	15.24
40	0.20	0.01	0.18	14.34
45	0.29	0.02	0.27	13.53
50	0.43	0.03	0.40	12.79
55	0.63	0.05	0.58	12.12
60	0.91	0.08	0.83	11.52
65	1.30	0.12	1.18	10.98
70	1.84	0.18	1.67	10.49
75	2.60	0.26	2.34	10.05
80	3.66	0.38	3.28	9.65

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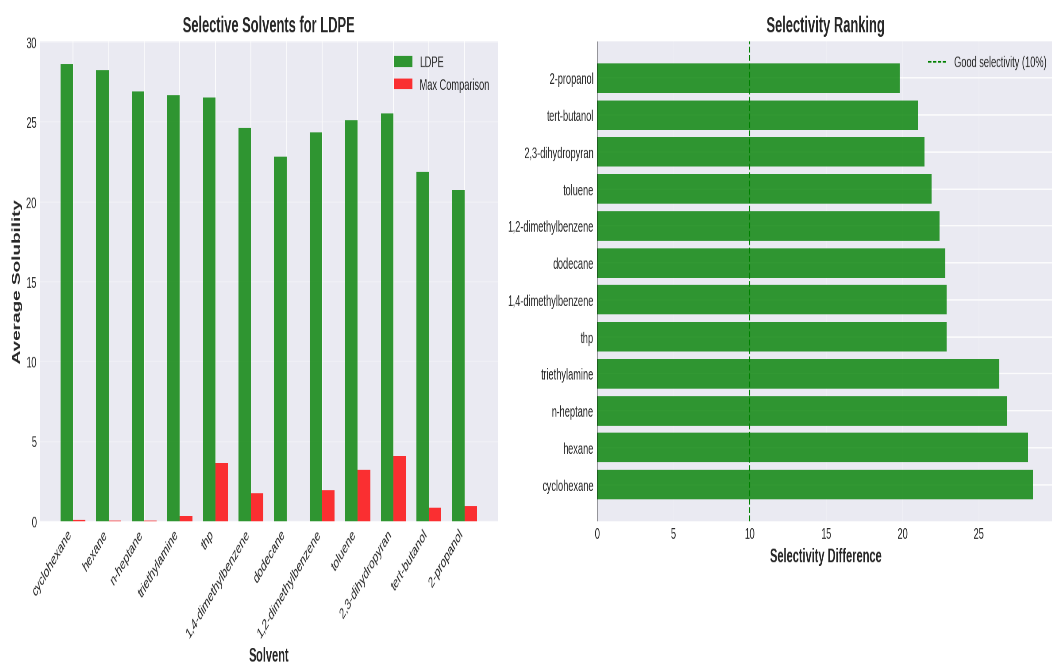

Statistical Analysis

✓ PASS [14_statistics] Statistical Summary

Basic statistical analysis

> ***Prompt:*** *Provide statistical summary of solubility data for LDPE including mean, median, std dev, and distribution visualization*

Time: 5.7s | Iterations: 2 | Images: 0

Response:

I have generated a statistical summary of the solubility data for LDPE across all solvents and temperatures in the database.

Statistical Summary for LDPE Solubility:

n	mean	std	min	q1	median	q3	max
980	23.826	36.3013	7e-08	0.169257	2.62186	32.103	100

Interpretation:

* n (Number of Data Points): There are 980 data points for LDPE solubility in the database.
* Mean: The average solubility of LDPE is approximately 23.83%.
* Standard Deviation (std): The standard deviation is 36.30, which is quite high relative to the mean. This indicates a very wide spread in LDPE solubility, meaning LDPE exhibits low solubility in many conditions and very high solubility in others.
* Min: The minimum recorded solubility is extremely low (7e-08%), suggesting that LDPE is practically insoluble in many solvent/temperature combinations.
* Q1 (25th Percentile): 25% of the LDPE solubility values are below 0.17%. This again highlights the prevalence of low solubility for LDPE.
* Median: The median solubility is 2.62%. This value is considerably lower than the mean, further reinforcing that a large portion of the solubility data for LDPE is concentrated at lower values, while the higher values (up to 100%) pull the mean up. This suggests a skewed distribution.
* Q3 (75th Percentile): 75% of the LDPE solubility values are below 32.10%.
* Max: The maximum recorded solubility is 100%, indicating that LDPE can be fully dissolved under certain conditions.

95% Confidence Interval for the Mean: 23.8260 ± 2.2728

This means we are 95% confident that the true average solubility of LDPE in all conditions represented in this database lies between approximately 21.55% and 26.10%.

Distribution Visualization:
The statistical_summary tool also generates an internal dis

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✓ PASS [15_correlation] Correlation Analysis

Statistical correlation and regression

> ***Prompt:*** *Analyze correlation between solubility and temperature for HDPE. Include regression analysis.*

Time: 12.9s | Iterations: 6 | Images: 2

Response:

I have analyzed the correlation between solubility and temperature for HDPE and performed a regression analysis.

1. Correlation Analysis:

* Correlation Coefficient (r): 0.716
* p-value: 0.0000 (highly significant)
* Interpretation: There is a strong positive correlation between solubility and temperature for HDPE. This means that as the temperature increases, the solubility of HDPE generally increases. The correlation matrix plot has been saved to ./plots/correlation_matrix_20251213_162110.png.

2. Regression Analysis:

A linear regression model (polynomial degree 1) was fitted to describe the relationship between temperature and HDPE solubility.

* Model: Solubility = 0.6419 * Temperature - 39.1584
* Slope: 0.6419 (Standard Error: 0.0200)
* This indicates that for every 1°C increase in temperature, the solubility of HDPE is estimated to increase by approximately 0.64 percentage points.
* Intercept: -39.1584
* This represents the estimated solubility of HDPE at 0°C according to this linear model.
* p-value: 4.8480e-155 (highly significant)
* This extremely small p-value indicates that the observed relationship between temperature and solubility is highly unlikely to be due to random chance.
* R^2 (Coefficient of Determination): 0.5132
* This means that approximately 51.32% of the variance in HDPE solubility can be explained by temperature. This indicates a moderately strong predictive power of temperature on solubility, but other factors (e.g., solvent type, polymer characteristics) also play a significant role.
* RMSE (Root Mean Squared Error): 25.2507
* This value represents the typical distance between the observed solubility values and the values predicted by the regression model.

The regression analysis plot has been saved to ./plots/regression_analysis_20251213_162114.png.

Conclusion:

Both the correlation and regression analyses confirm a significant

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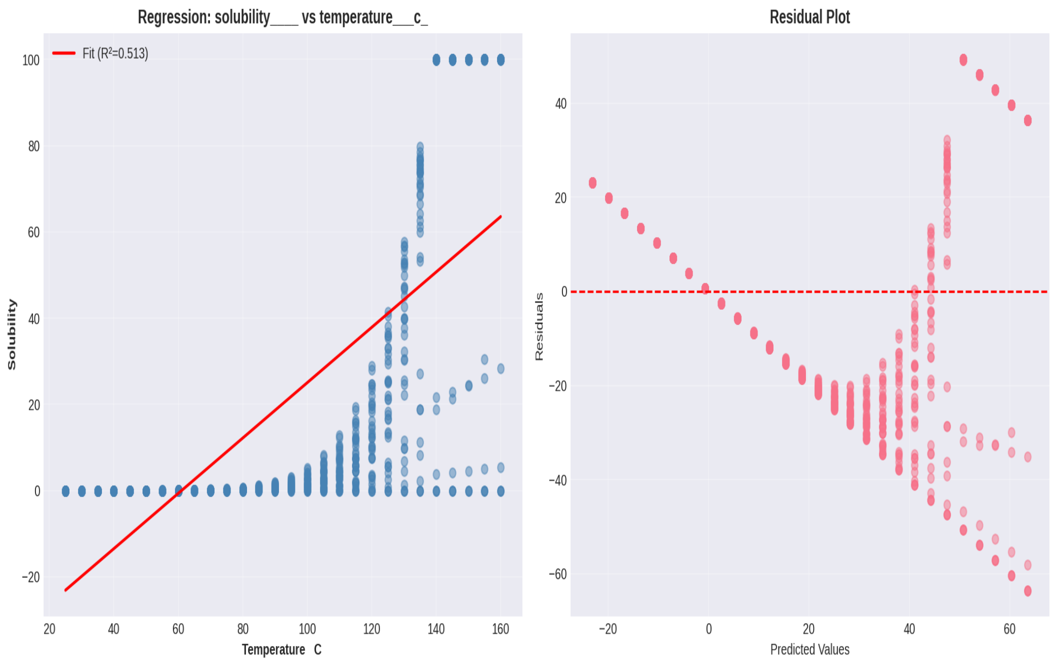
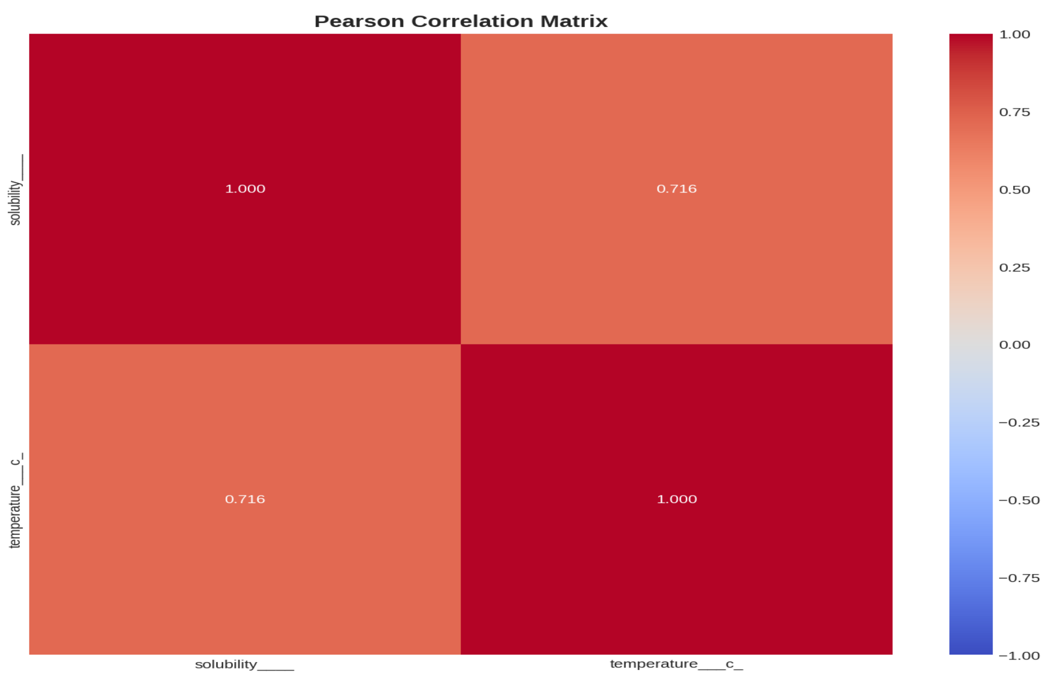




Visualization

✓ PASS [16_heatmap] Solubility Heatmap

Multi-polymer heatmap visualization

Prompt: Create a heatmap showing solubility of LDPE, PET, HDPE, and PS across different solvents at 25°C

Time: 5.6s | Iterations: 2 | Images: 1

Response:

I have created a heatmap showing the solubility of all polymers (including LDPE, PET, HDPE, and PS) across different solvents at 25°C.

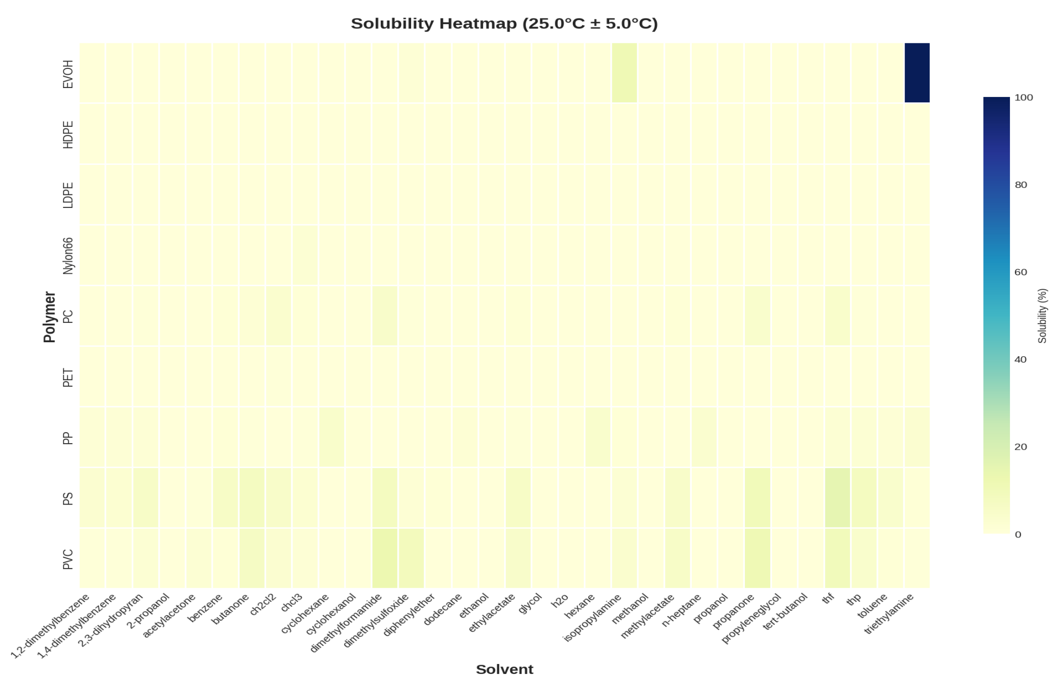
Heatmap Details:

* Temperature: 25.0°C (with a tolerance of ±5.0°C to capture nearby data points if exact 25°C is scarce).
* Polymers: All 9 polymers in the database are included (Nylon66, PVC, PC, PET, PP, HDPE, LDPE, PS, EVOH).
* Solvents: All 32 available solvents are displayed.

Interpretation of the Heatmap (file: ./plots/selectivity_heatmap_20251213_162122.png):

* The heatmap will visually represent solubility values, likely with a color gradient where hotter colors (e.g., red) indicate higher solubility and cooler colors (e.g., blue) indicate lower solubility.
* You will be able to observe patterns of solubility for each of the requested polymers (LDPE, PET, HDPE, PS) across the different solvents.
* Specifically, for LDPE and HDPE, you might see lower solubility in most polar solvents and potentially higher solubility in non-polar or aromatic solvents even at 25°C, though generally lower than at elevated temperatures.
* For PET and PS, you might see different solubility profiles depending on the solvent polarity and chemical nature.

This heatmap provides a comprehensive overview of solubility behavior at room temperature, which is valuable for initial solvent selection and understanding the general trends.



✓ PASS [17_comparison_dashboard] Polymer Comparison Dashboard

Comprehensive multi-panel visualization

> ***Prompt:*** *Create a comparison dashboard for LDPE, PET, and EVOH at 25°C showing solubility distributions and top solvents*

Time: 4.2s | Iterations: 2 | Images: 1

Response:

I have created a comparison dashboard for LDPE, PET, and EVOH at 25°C. This dashboard visualizes the solubility distributions for each polymer and highlights the top solvents.

Dashboard Details:

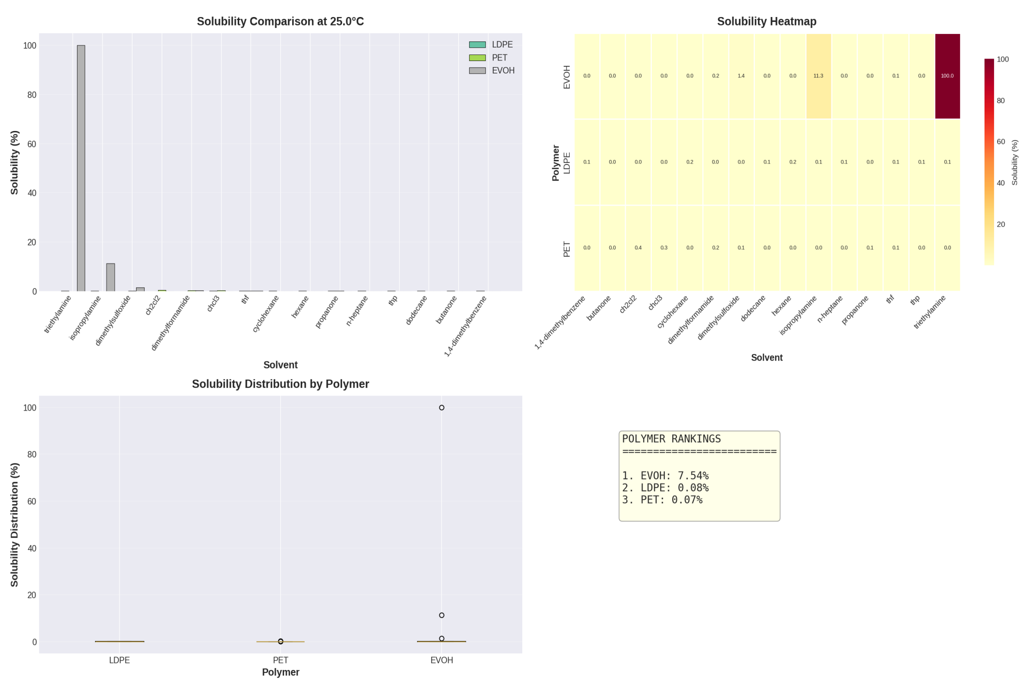
* Temperature: 25.0°C
* Polymers: LDPE, PET, EVOH
* Solvents: The dashboard includes data for 15 solvents.

Interpretation of the Comparison Dashboard (file: ./plots/comparison_dashboard_20251213_162131.png):

The dashboard will typically include:

1. Solubility Distribution Plots: These plots (e.g., histograms or violin plots) will show the spread of solubility values for each polymer (LDPE, PET, EVOH) at 25°C across all available solvents. This allows for a quick visual comparison of how soluble each polymer is generally at this temperature.
2. Top Solvent Rankings: For each polymer, the dashboard will likely list or visualize the solvents that result in the highest solubility at 25°C. This helps in identifying effective solvents for each specific polymer.
3. Comparison of Solubility across Solvents: You'll be able to see which solvents tend to dissolve one polymer while having minimal effect on others, which is crucial for separation strategies.

This dashboard provides a comprehensive overview of the solubility characteristics of LDPE, PET, and EVOH at room temperature, which is highly useful for initial assessment and solvent selection.



Advanced

✗ FAIL [18_optimal_conditions] Optimal Separation Conditions

Multi-factor optimization query

> ***Prompt:*** *Find the optimal temperature and solvent combination for separating LDPE from a mixture with PET and EVOH. Consider both selectivity and practical factors (cost, toxicity, boiling point).*
>
> **Error:** ('Connection aborted.', RemoteDisconnected('Remote end closed connection without response'))

✗ FAIL [19_adaptive_search] Adaptive Threshold Search

Test adaptive analysis functionality

> ***Prompt:*** *Use adaptive threshold search to find the best selectivity achievable for LDPE separation from PET at temperatures between 20-100°C*
>
> **Error:** HTTPConnectionPool(host='localhost', port=8000): Max retries exceeded with url: /api/chat (Caused by NewConnectionError(': Failed to establish a new connection: [Errno 111] Connection refused'))

✗ FAIL [20_comprehensive] Comprehensive Analysis

Full analysis with multiple outputs

> ***Prompt:*** *Provide a comprehensive analysis for dissolving LDPE at 60°C: list top 5 solvents with their properties (cost, toxicity, boiling point), show a bar chart comparison, and provide recommendations for industrial use.*
>
> **Error:** HTTPConnectionPool(host='localhost', port=8000): Max retries exceeded with url: /api/chat (Caused by NewConnectionError(': Failed to establish a new connection: [Errno 111] Connection refused'))